

PROJECT STATUS

100X Improvement in Bioelectronic Signal Acquisition

Resonant electrophysiology - coupled LC resonators with Q-enhancement

UCLA (Prof. Tyler Clites / Samantha Herman) - UCI (Prof. C. Velez Cuervo / Juan Diego Sanchez)

19 August 2026

Where the project stands

The physics works. A two-channel design point is verified end to end in simulation: co-tuned notches at 55.89 and 100.00 MHz, both meeting the bandwidth and loaded-Q specification on a ground-free differential line. That is the central claim of the project and it holds.

What does not work yet is everything between that result and a device. This session put measured IGZO TFT data into the picture for the first time, and it changed the outlook: the transistors assumed in every simulation to date are roughly 1400x stronger than the ones actually fabricated. At 1.5 V the measured devices fall 9x to 64x short of the transconductance the resonator needs, and no amount of bias current closes that gap, because the supply caps the gate drive.

Separately, the channel architecture had a structural flaw that is now understood and fixed on paper: sharing the return path moved resonances 27-38 % off their design frequencies, and got worse with every channel added. Giving each channel its own return holds the error under 2.4 %.

One problem solved, one hard problem newly quantified, and a decision now on the table about supply voltage and device technology.

2413 / 7217

loaded Q achieved
(spec: ≥ 1667)

9x - 64x

transconductance
shortfall at 1.5 V

2.4% -> 1.2%

channel placement error
with isolated return

Requirements scorecard

Every system requirement against what has actually been demonstrated

REQUIREMENT	TARGET	STATUS TODAY	VERDICT
Operating band	50 - 200 MHz	55.89 and 100.00 MHz verified	MET
Notch bandwidth	≤ 30 kHz	23.2 kHz / 13.9 kHz	MET
Loaded Q	≥ 1667 @ 50 MHz	Q_L 2413 / 7217	MET
Ground-free operation	mandatory	differential, no ground node	MET
Channel count	≥ 1000	10 placed correctly (isolated ret.)	PARTIAL
Power per channel	≤ 100 μ W	3.5 mW ideal device; worse with TFT	NOT MET
Real devices (IGZO)	gm 0.8 - 2.0 mS	89 μ S max at VDD = 1.5 V	NOT MET
Bias stability	± 0.25 % on Itail	open loop only, no AGC	NOT MET
Footprint	150 x 150 μ m	not laid out yet	NOT STARTED
EMG signal path	± 5 mV, 20 Hz-2 kHz	not simulated yet	NOT STARTED
Inductor Q0	assumed 31	not measured on film	NOT STARTED

How to read this

MET means demonstrated in closed-loop simulation, not calculated. The four met requirements were re-verified this session against a freshly generated netlist, reproducing 55.8901 MHz / 35.90 dB / 22.98 kHz / Q_L 2432 and 100.0021 MHz / 39.69 dB / 13.98 kHz / Q_L 7151.

PARTIAL on channel count: ten channels were placed at their design frequencies with the isolated return, which is the mechanism that scales. Nothing above ten has been simulated, and 1000 is still far away.

NOT MET means measured and found short, with a number attached.

NOT STARTED means no work has been done - not that it is expected to fail.

100X Improvement in Bioelectronic Signal Acquisition - project status

What has been achieved

Results that are verified, not assumed

■ A verified two-channel design point

Two co-tuned notches on one ground-free differential line:

channel A	55.890 MHz	35.9 dB deep	23.2 kHz BW	Q_L 2413
channel B	100.002 MHz	39.7 dB deep	13.9 kHz BW	Q_L 7217

Both meet the ≤ 30 kHz bandwidth spec and beat the $Q_L \geq 1667$ target.

Reproduced this session from a newly written netlist generator, so the result does not depend on any one hand-edited file.

■ An open, reproducible simulation flow

xschem + ngspice, no proprietary tools, running end to end.

Readers for the lab's raw .xls and .xlsx data written from scratch (no xlr/openpyxl available), cross-checked against the pre-existing analysis: 172 of 172 saturation-current values reproduced exactly.

■ The first real device model

IGZO TFT parameters extracted from the INRF measurements by two independent routes that agree to 0.5 % on long-channel devices:

$K_p = 0.72 \text{ uA/V}^2$ $V_{th} \sim 0.1 \text{ V}$ $\lambda \sim 0.007$ (best chip)

Written up as test1/models/tft_igzo_test1.lib with four variants spanning best case, typical, short-channel and worst-chip.

■ A structural fix for scaling

The shared-return flaw is identified, quantified and solved on paper. Isolated return holds channel placement to ≤ 2.4 % error, improving to 1.2 % at ten channels, where the shared return degrades to 37.9 %.

■ Quantified design levers

gm required / tank loss = 854 uS/Ohm, constant to ± 4 % over a 16x range of series resistance. This converts any future inductor-quality result directly into a transconductance requirement.

Contact-limited length $L_c \sim 54 \text{ um}$, so the contact problem is bounded and its payoff is known: 5.6x in K_p , 2.4x in gm.

Problem 1 - The TFT cannot reach the required gm at 1.5 V

BLOCKING

What we found

The tank needs $g_m = 0.81 \text{ mS}$ (100 MHz) to 2.02 mS (55.9 MHz).

The best measured device gives 89 uS at $V_{DD} = 1.5 \text{ V}$; the realistic short-channel device gives 32 uS . Shortfall: 9x to 64x.

Raising the bias current does not help - V_{GS} is capped by the rail.

$K_p = 0.72 \text{ uA/V}^2$ measured vs 1000 uA/V^2 assumed (factor ~ 1400).

Possible solutions

Raise the supply

IGZO TFTs normally run at 5-20 V. Channel B closes at $V_{DD} \sim 13 \text{ V}$, channel A at $\sim 32 \text{ V}$. Cheapest fix technically, but it breaks the power budget and the implant safety case - needs a decision.

Increase Cox

$K_p = \mu \cdot C_{ox}$ and μ is fixed by the material. A thinner or higher-k gate dielectric raises K_p proportionally. Going from a $\sim 100 \text{ nm}$ SiO_2 -equivalent to a high-k stack is worth 10-30x.

Fix the contacts

Worth 5.6x in K_p on short-channel devices for free (see problem 4). Not sufficient alone, but multiplies with the others.

Change the active device

If the rail must stay at 1.5 V , no TFT technology reaches this g_m . That is a topology-level decision.

Problem 2 - The shared return destroys the channel frequency plan

SOLVED

What we found

All channels returned to VDD through the same node, so the N return inductors sat in parallel: effective return = $(L/2)/N$, shrinking with N . Modes landed 27-38 % off their design frequencies, and the error grew with channel count. Channel 10 landed at 275.8 MHz - out of band.

Confirmed by simulation at $N = 2, 3, 4, 6, 8, 10$.

Possible solutions

Isolated return per channel

Each channel gets its own floating return, coupled to the line at both ends so the design stays ground-free. Placement error drops to $\leq 2.4\%$, and it improves with N (1.2 % at $N=10$). Implemented in `scale/gen_nchan.py --return isolated`.

Cost

One extra inductor and one extra coupling capacitor per channel. The area cost is real and must be checked against the 150 x 150 μm cell.

Problem 3 - Bias sensitivity with no amplitude control

BLOCKING

What we found

Q-enhancement works by cancelling tank loss, so the operating point sits just below oscillation. Measured previously: $\pm 0.25\%$ error on I_{tail} still gives ≥ 17 dB depth, but $+1\%$ on channel B makes it oscillate. No feedback loop exists - everything so far is open loop.

Intrinsic to negative-gm Q-enhancement; consistent with the literature.

Possible solutions

Amplitude control loop (AGC)

Detect the resonator amplitude and servo the tail current. This is the standard answer and the single largest piece of unbuilt work.

Isolated return helps

With channels no longer shifting each other's thresholds, each loop can be designed and tuned independently instead of co-tuned as a coupled system.

Note

This could not be re-measured this session: AC analysis is linear and cannot detect oscillation. Needs pole-zero (.pz) or transient analysis.

Problem 4 - Contact resistance dominates the short-channel device

HIGH

What we found

K_p is flat at $\sim 0.72 \text{ uA/V}^2$ for $L \geq 40 \text{ um}$ and collapses to 0.13 uA/V^2 at $L = 8 \text{ um}$. Linear and saturation extractions agree to 0.5 % on long channels and diverge to a ratio of 0.64 on short ones - the contact fingerprint.
 $2 \cdot R_c \cdot W = 6.6 \text{e6 Ohm} \cdot \text{um}$ ($660 \text{ Ohm} \cdot \text{cm}$); contact-limited length $L_c \sim 54 \text{ um}$.

$\sim 100x$ worse than good IGZO practice - a process problem, not a material one.

Possible solutions

Contact engineering

Self-aligned source/drain, an n+ interlayer, or a post-metallisation anneal.
Standard IGZO practice recovers 1-2 orders.

Payoff

Recovering the intrinsic 0.72 uA/V^2 at $L = 8 \text{ um}$ is 5.6x in K_p , i.e. 2.4x in gm, for no extra area and no extra power.

Do this before fabricating more

It changes the device model that every downstream design decision depends on.

Problem 5 - Power budget is orders of magnitude over

HIGH

What we found

Budget is 100 uW/channel (100 mW for 1000 channels). The verified two-channel point draws 7.1 mW - 3.5 mW/channel, 35x over, and that is with the optimistic device. With the measured TFT the required current is far higher.

Bias current scales as $gm^2/(Kp(W/L))$; a low Kp is paid for in current.*

Possible solutions

Duty-cycling

Channels do not all need to be live at once. EMG bandwidth is 2 kHz against a 150 kHz channel allocation - a large duty-cycle margin.

Lower the gm requirement

$gm_{req} / RS = 854 \text{ uS/Ohm}$ was measured, constant to $\pm 4\%$ over a 16x range of RS. Lower tank loss lowers the required gm proportionally - but thin-film inductors go the wrong way (see 7).

Device sizing

Larger W/L buys gm at constant current, bounded by area.

Problem 6 - Measurements that are missing outright

MEDIUM

What we found

VGS/, COX/ and HALL/ are empty for all three chips.

The CAP test structure area is not recorded, so Cox cannot be normalised.

W and L are not recorded for the devices in TFT/test/ - which hold the best transfer sweeps in the whole set and currently cannot be used quantitatively.

Without Cox we cannot separate μ from Cox, nor predict gate capacitance.

Possible solutions

Measure Cox on a known area

A single CV sweep on a pad of recorded dimensions closes this. It also tells us whether the low Kp is thick dielectric or low mobility - which decides whether problem 1 is fixable.

Record geometry with the data

Add W, L and pad spacing to the Clarius test names, so extraction stays automatic.

Gate capacitance matters here

At 50-200 MHz the gate capacitance loads the tank directly. It is currently a placeholder in the model.

Problem 7 - Unverified assumptions still carried in the design

MEDIUM

What we found

Inductor $Q_0 = 31$ is assumed, never measured on an actual thin film.

Realistic film inductors are usually worse, which raises the required gm in direct proportion (854 $\mu\text{S}/\Omega$ measured).

The 150 x 150 μm footprint has never been laid out.

These feed straight into problems 1 and 5 and could make both worse.

Possible solutions

Measure or simulate one film inductor

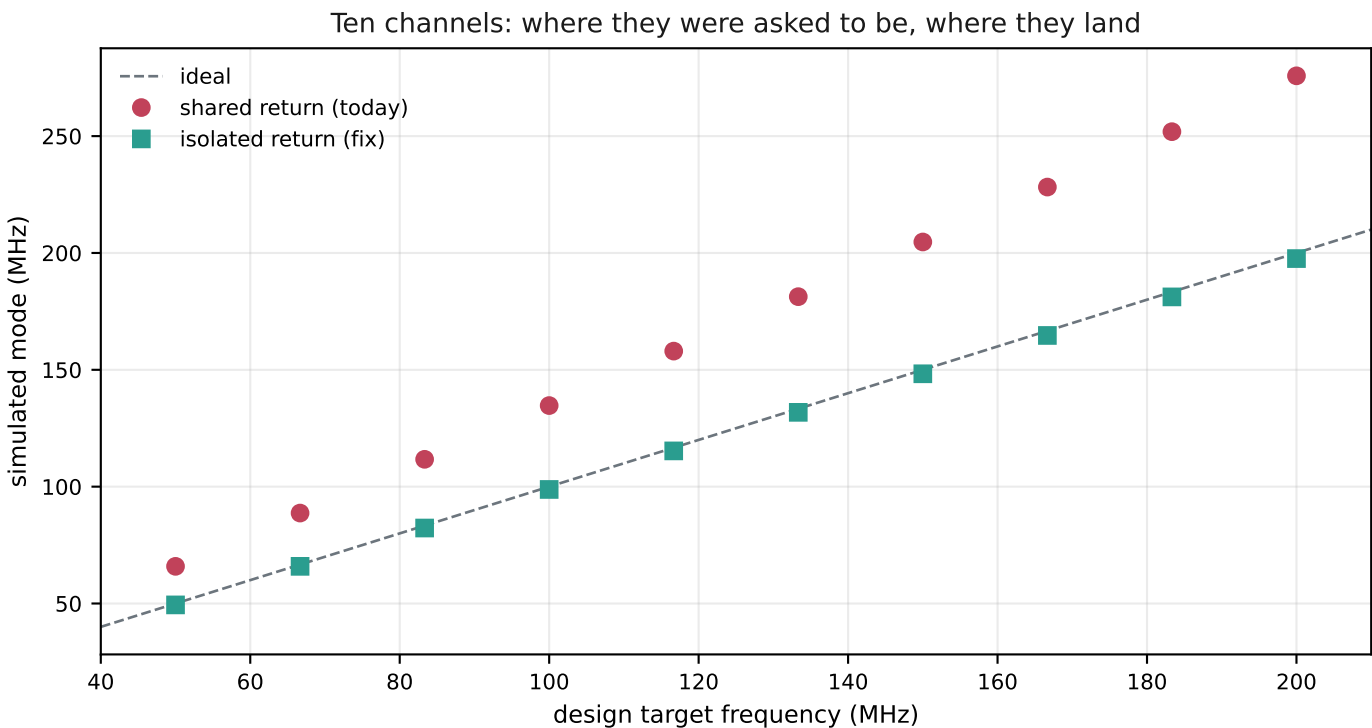
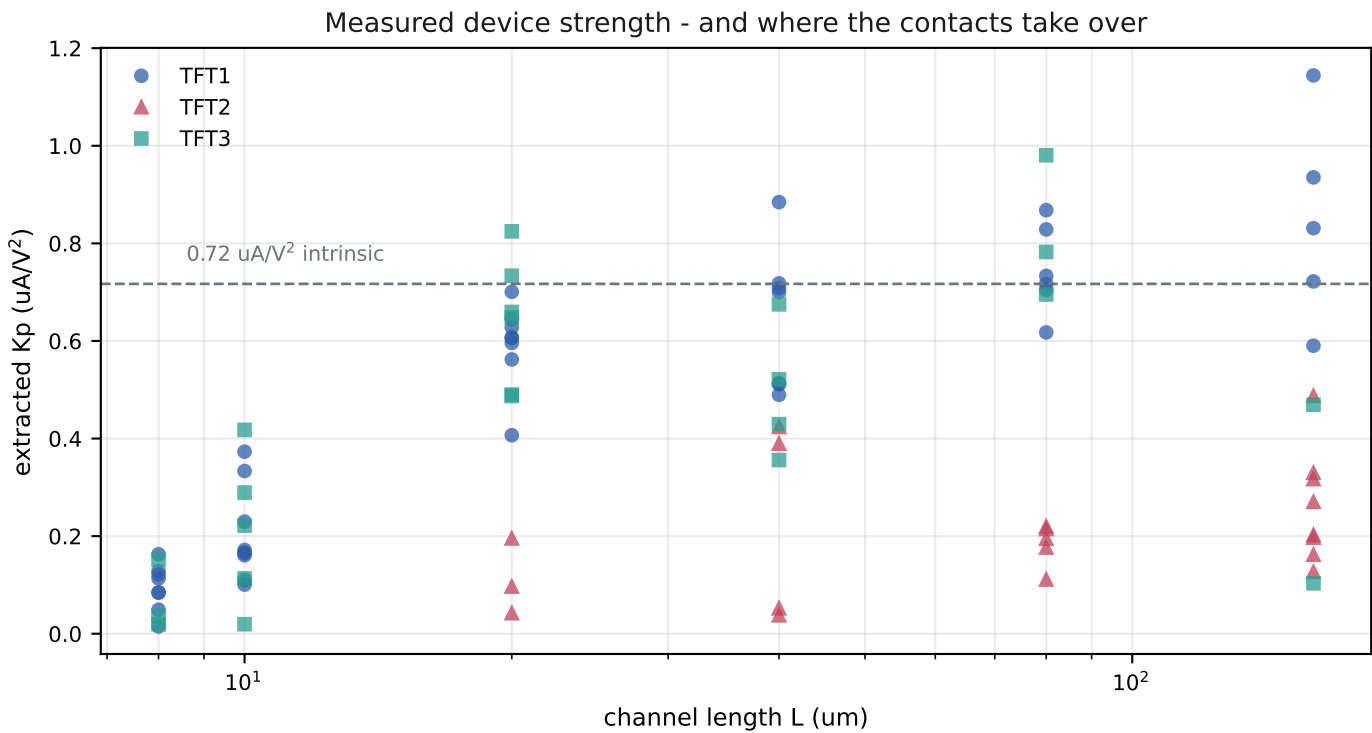
Cheap, and it either confirms or invalidates the loss budget the whole design rests on.

Draw one channel

A single cell layout answers the footprint question and prices the extra inductor the isolated return needs.

Evidence behind the two hard numbers

The transconductance shortfall and the scaling fix



Verification performed this session

Raw .xls reader vs the pre-existing Isat reports: 172 / 172 values exact.
Transfer-curve extraction at $V_{DS} = 0.1 / 0.1 / 1 / 2 \text{ V}$ gave 26.4 / 27.0 / 27.1 / 28.4 $\mu\text{A}/\text{V}^2$ - 7 % spread across a 20x range of drain voltage - and the extracted threshold reproduces the instrument's own V_T column.
The netlist generator reproduces the verified two-channel design point before it was used for anything else.

100X Improvement in Bioelectronic Signal Acquisition - project status
Not settled: channel-to-channel interaction could not be re-measured,
because AC analysis is linear and cannot detect oscillation. The 9 % figure

What is missing, and in what order

Sequenced so that the decisions that can invalidate work come first

Now

- Adopt the isolated return in all further work (scale/gen_nchan.py).
- Measure Cox on a pad of known area; record W/L with every device.
- Re-derive the channel interaction with .pz or transient, not AC.

Next

- Decide the supply-voltage question - it is a project-level call, not a tweak.
- Contact engineering trial, then re-extract the model.
- Measure or simulate one realistic thin-film inductor (Q0).

Then

- Design the amplitude control loop once the real gm is known.
- Lay out one channel and price the isolated-return area cost.
- Simulate the EMG signal path end to end (+-5 mV, 20 Hz - 2 kHz).

The decision that gates everything else

The 1.5 V supply was inherited from a silicon-CMOS assumption, not from a requirement of the application. Every TFT technology - not just this one - will struggle to deliver milliamp-scale transconductance from that rail.

Three paths exist and they lead to different projects:

- 1. Raise the supply, and re-open the power and safety budgets.**
- 2. Raise Cox with a high-k or thinner gate stack, and keep 1.5 V.**
- 3. Keep 1.5 V and change the active device away from a TFT.**

Nothing downstream - AGC design, power budget, layout - can be settled before this is. It should be taken deliberately, with UCLA in the room, rather than defaulted into.

*Detail behind every number in this report: [report/device_and_scaling_findings.pdf](#)
Extraction and simulation code: [tools/](#), [tft/](#), [scale/](#)*